Chemistry • Reversible reactions and Le Chatelier's principle

Equilibria

Official exercise • Complete solved answers • Dynamic-equilibrium revision

Chapter 6 • Complete official exercise • 10 MCQs + 5 short + 5 constructed + 5 descriptive + 2 investigative

What this chapter is about

• **Dynamic equilibrium** occurs in a closed reversible system when forward and reverse rates become equal. Both reactions continue, but concentrations remain constant.
• **Le Chatelier's principle** predicts the direction of disturbance: concentration, gas pressure and temperature changes can shift the equilibrium position.
• **A catalyst** speeds both directions equally, so equilibrium is reached sooner but its position and equilibrium constant do not change.
• **Industrial yield** can be improved by suitable conditions, continuous product removal and recycling unreacted reactants, as in the Haber process.

Exercise coverage: All five official sections are solved in book order, including both investigative tasks.

A. Multiple Choice Questions

1. What will happen if the rates of forward and reverse reactions are very high?

(a) the equilibrium point will be reached very soon
(b) the equilibrium point will be reached very late
(c) the reaction will not attain the state of dynamic equilibrium
(d) the reaction will be practically irreversible

Answer: (a) the equilibrium point will be reached very soon — High forward and reverse rates allow the system to reach the equal-rate dynamic state more quickly.

2. Predict which components of the atmosphere react in the presence of lightning.

(a) N_2 and H_2O
(b) O_2 and H_2O
(c) CO_2 and O_2
(d) N_2 and O_2

Answer: (d) N_2 and O_2 — Lightning supplies enough energy for $N_2 + O_2 \rightarrow 2NO$ despite nitrogen's strong triple bond.

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3. One mole of PCl_5 is placed in container A and one mole each of Cl_2 and PCl_3 in container B. Both are sealed and heated to the same temperature to reach equilibrium. Predict the compositions.

(a) both containers will have the same composition
(b) container A will have more PCl_3 than B
(c) container A will have less PCl_3 than B
(d) both containers will have zero concentration of reactants

Answer: (a) both containers will have the same composition — For $PCl_5 \rightleftharpoons PCl_3 + Cl_2$, these are equivalent stoichiometric starting points approached from opposite directions. At the same temperature—and assuming identical volumes—they reach the same equilibrium composition.

4. CaO is produced by reversible decomposition of $CaCO_3$. Choose conditions to produce the maximum amount of lime.

(a) heating at high temperature in a closed vessel
(b) heating at high temperature in an open vessel
(c) cooling in a closed vessel
(d) cooling in an open vessel

Answer: (b) heating at high temperature in an open vessel — High temperature favours the endothermic decomposition, and an open vessel lets CO_2 escape, driving $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$ forward. The stem's word "exothermic" is a source error.

5. What condition should be met for a reversible reaction to achieve equilibrium?

(a) all reactants should be converted into products
(b) 50% of reactants should be converted into products
(c) concentrations of all reactants and products should become constant
(d) one product should be removed from the reaction mixture

Answer: (c) concentrations of all reactants and products should become constant — At dynamic equilibrium the macroscopic concentrations remain constant because forward and reverse rates are equal, not because reaction stops.

6. Why does gas start coming out when you open a can of fizzy drink?

(a) because the solubility of the gas increases
(b) because the gas is insoluble in water
(c) because the gas is dissolved under pressure and comes out when pressure decreases
(d) because the solubility of the gas decreases at high pressure

Answer: (c) because the gas is dissolved under pressure and comes out when pressure decreases — Opening lowers the pressure of CO_2 above the drink. Its solubility falls, so dissolved CO_2 escapes as bubbles.

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7. $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$ is performed in a closed vessel. How is equilibrium affected if pressure is increased?

(a) the forward reaction is favoured
(b) the backward reaction is favoured
(c) no effect on backward reaction
(d) no effect on forward or backward reaction

Answer: (b) the backward reaction is favoured — Higher pressure favours the side with fewer gas moles. The reactant side has no gaseous species, so CO_2 combines with CaO and the backward reaction is favoured.

8. When will a reaction become reversible?

(a) if activation energy of the forward reaction is comparable to that of the backward reaction
(b) if forward activation energy is higher than backward activation energy
(c) if forward activation energy is lower than backward activation energy
(d) if enthalpy change of both reactions is zero

Answer: (a) if activation energy of the forward reaction is comparable to that of the backward reaction — Comparable forward and reverse barriers allow both directions to proceed at appreciable rates under the same conditions.

9. Is a reversible reaction useful for preparing compounds on a large scale?

(a) no
(b) yes
(c) useful only when equilibrium lies far to the right
(d) useful only when equilibrium lies far to the left

Answer: (b) yes — Yes. Industrial reversible reactions such as ammonia synthesis give useful products when conditions, product removal and recycling are controlled.

10. What happens to product concentrations if a reversible reaction at equilibrium is not disturbed?

(a) they remain constant
(b) they keep increasing
(c) they keep decreasing
(d) they remain constant for some time and then start decreasing

Answer: (a) they remain constant — Undisturbed dynamic equilibrium has constant macroscopic concentrations while both reactions continue at equal rates.

B. Questions for Short Answers

6.1. How is dynamic equilibrium different from static equilibrium?

In **dynamic equilibrium**, opposing microscopic processes continue at equal nonzero rates, so macroscopic properties stay constant; for example, reactants form products as fast as products reform reactants. In **static equilibrium**, there is no continuing opposing process or motion, such as a stationary object with balanced forces.

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6.2. How will increasing temperature affect $2H_2O(l) \rightleftharpoons 2H_2(g) + O_2(g)$, when energy/electricity is supplied?

Decomposition of water is **endothermic**. Treat the supplied energy as favouring the forward direction. Increasing temperature therefore favours the endothermic forward reaction and shifts equilibrium towards H_2 and O_2, increasing the equilibrium constant for the decomposition.

Electrolysis is driven by electrical energy rather than by a simple spontaneous thermal equilibrium, but the temperature rule for the endothermic direction remains the same.

6.3. How can you get the maximum yield in a reversible reaction?

Apply Le Chatelier's principle to that specific equation: add reactant, remove product continuously, choose pressure to favour fewer gas moles, and choose temperature to favour the thermally favourable direction while maintaining a practical rate. A catalyst shortens waiting time but does not itself increase equilibrium yield.

6.4. How can you decrease the time to attain equilibrium in a reversible reaction?

Add a suitable catalyst to lower activation energy for both directions, increase temperature when compatible with the desired yield, and use adequate mixing and reactant concentrations or gas pressure. The catalyst increases both rates equally and changes only the time required, not the final equilibrium composition.

6.5. What is the effect of increasing pressure on $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$ at equilibrium?

There are two moles of gas on each side, so $\Delta n_{gas} = 0$. Increasing pressure by reducing volume therefore has **no effect on equilibrium position**. Both concentrations and rates initially increase, but no directional shift is needed.

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C. Constructed Response Questions

6.1. Why are some reactions irreversible while others are reversible?

A reaction appears reversible when products can reform reactants at an appreciable rate under the same conditions. This is favoured when forward and reverse activation barriers are comparable and products are not vastly more stable than reactants.

A reaction is practically irreversible when its reverse rate is negligible: products may be much more stable, the reverse activation energy may be enormous, or a product may escape or undergo another reaction.

Thus reversibility depends on **energy differences, activation barriers and physical conditions**. "Irreversible" often means irreversible for practical purposes, not that the reverse process is absolutely impossible.

6.2. Why are combustion reactions generally irreversible?

Combustion is strongly exothermic and forms very stable products such as CO_2 and H_2O. Recreating the fuel and oxygen requires an extremely large energy input and has a prohibitively high activation barrier under ordinary conditions. In an open system, gaseous products also disperse, preventing a measurable reverse reaction.

6.3. Can you make an irreversible reaction reversible and vice versa?

Conditions can change the **practical** reversibility of some reactions, but not every irreversible reaction becomes usefully reversible.

• **Reversible to effectively complete:** remove a product as gas, precipitate or distillate, or add excess reactant so equilibrium continually shifts forward.

• **Apparent irreversible to reversible:** seal the vessel to retain products and change temperature, pressure or concentrations so that products can react backward.

• **Example:** $CaCO_3 \rightleftharpoons CaO + CO_2$ reaches equilibrium in a closed vessel but decomposes nearly completely in an open kiln as CO_2 escapes.

Strongly irreversible processes such as ordinary combustion cannot be made practical reversible laboratory preparations merely by sealing a container.

6.4. How do you know if a reaction is reversible or irreversible?

Test whether products can reform reactants under the same or suitable conditions. In a closed system, a reversible reaction reaches constant macroscopic properties with reactants and products still present; starting from either side can approach the same equilibrium composition. Isotopic tracers can show continued exchange even at equilibrium. A practically irreversible reaction shows no measurable reverse reaction and usually proceeds until a limiting reactant is consumed.

6.5. Are phase changes in water (solid to liquid, liquid to vapour) reversible or irreversible?

They are **reversible physical changes**:

$$H_2O(s) \rightleftharpoons H_2O(l) \text{ melting / freezing}$$

$$H_2O(l) \rightleftharpoons H_2O(g) \text{ vaporization / condensation}$$

At a suitable temperature and pressure in a closed system, opposing phase changes can occur at equal rates, producing dynamic equilibrium. Water’s chemical identity and H_2O molecules remain unchanged.

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D. Descriptive Questions

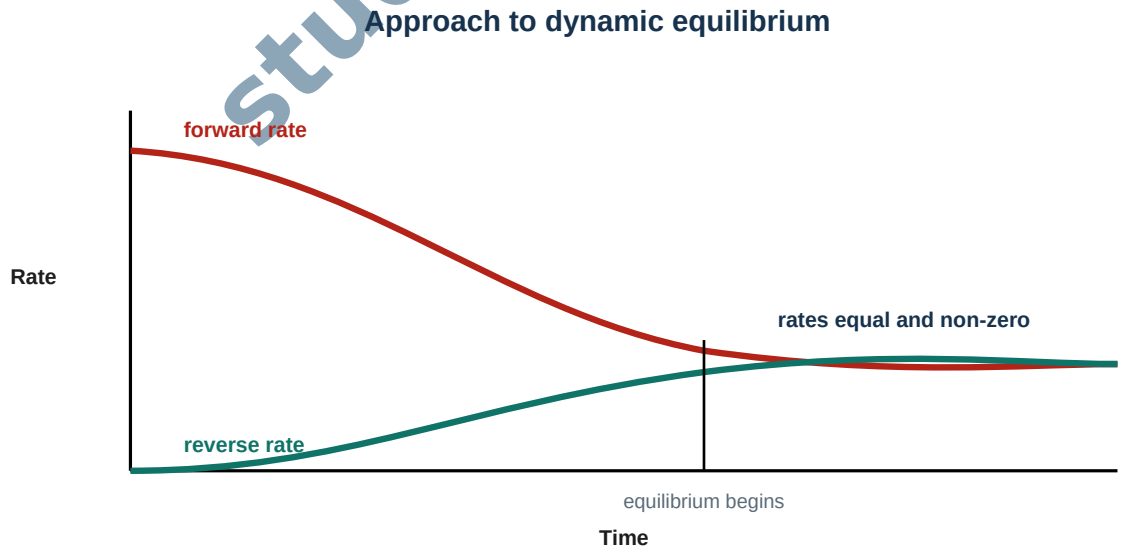
6.1. How can you drive a reversible reaction at equilibrium (a) forward and (b) backward?

Change	Drive forward (towards products)	Drive backward (towards reactants)
Reactant concentration	Add a reactant	Remove a reactant
Product concentration	Remove a product continuously	Add a product
Gas pressure	Favour forward if products have fewer gas moles	Favour backward if reactants have fewer gas moles
Temperature	Raise T if forward is endothermic; lower T if forward is exothermic	Raise T if backward is endothermic; lower T if backward is exothermic
Physical separation	Distil, precipitate or vent product	Condense or otherwise remove a gaseous reactant where feasible

A catalyst cannot select either direction. Actual industrial choices must also consider rate, safety, cost and energy use.

6.2. Explain how the rates of forward and backward reactions change when a system approaches equilibrium.

Figure 1 — Original rate-time graph for a reversible reaction started with reactants



After equilibrium: both reactions continue, but concentrations remain constant because their rates match.

Initially, the forward rate is high because reactant concentration is high, while the reverse rate is zero or very small because little product is present. As reactants are consumed, the forward rate decreases. As products accumulate, the reverse rate

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increases.

Eventually the two rates become **equal but not zero**. Concentrations then remain constant and dynamic equilibrium is established.

6.3. Describe the effect of a catalyst on a reversible reaction.

A catalyst provides an alternative lower-activation-energy pathway for both forward and reverse reactions. Both rates increase, so the system reaches equilibrium more quickly.

Quantity	Effect of catalyst
Forward activation energy	Decreases
Reverse activation energy	Decreases
Time to reach equilibrium	Decreases
Equilibrium position	No change
Equilibrium constant K	No change at fixed temperature
Enthalpy change ΔH	No change

Thus a catalyst improves **rate**, not equilibrium yield. Temperature remains the variable that changes K.

6.4. How can a reversible reaction be forced to go to completion?

Continuously disturb equilibrium in the forward direction so it can never settle with appreciable reactants remaining:

• **Remove a product:** allow a gas to escape, precipitate an insoluble product, distil a volatile product or absorb a product in another phase.

• **Keep adding reactant:** maintain a high reactant concentration and use excess of the less valuable reactant.

• **Choose favourable conditions:** adjust temperature and gas pressure to favour products, while retaining a practical rate.

• **Consume product in a following reaction:** a coupled step prevents its accumulation and reverse reaction.

Example: $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)\uparrow$ in an open kiln

In a sealed system without product removal, a genuinely reversible reaction does not normally reach 100% conversion.

6.5. How does change in temperature affect a reaction at equilibrium?

Temperature changes the value of the equilibrium constant and favours one direction according to the sign of ΔH.

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Temperature change	Favoured direction	Effect on K
Increase	Endothermic direction	K increases if forward is endothermic; decreases if forward is exothermic
Decrease	Exothermic direction	K decreases if forward is endothermic; increases if forward is exothermic

For example, in $N_2 + 3H_2 \rightleftharpoons 2NH_3$ (forward exothermic), cooling favours NH_3 but slows both rates. Heating increases rate but reduces equilibrium NH_3 yield; this is why a compromise temperature and catalyst are used industrially.

E. Investigative Questions

6.1. Study the effect of heat on hydrated copper sulphate. Why does this salt look coloured and why does it lose colour upon heating?

Blue copper sulphate crystals contain water of crystallization, $CuSO_4 \cdot 5H_2O$. Water molecules coordinated to Cu^{2+} produce a ligand environment that gives the hydrated copper(II) species its characteristic blue colour.

$$CuSO_4 \cdot 5H_2O(s) \rightleftharpoons CuSO_4(s) + 5H_2O(g)$$

blue hydrate ⇌ white/grey anhydrous salt + steam

Heating drives off water and changes the environment around Cu^{2+}; the anhydrous salt is white to greyish white. Adding water restores the blue hydrated species, demonstrating reversibility. Heating is required because dehydration is endothermic. Use eye protection and avoid inhaling dust or touching hot apparatus.

6.2. Ammonia synthesis is important for urea fertilizer. Explain the conditions for maximum ammonia yield.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)\ \Delta H \approx -92\ \text{kJ for 2 mol } NH_3$$

1. High pressure: Four gas moles form two, so high pressure favours NH_3. Industrial plants commonly use roughly 200 atm, balanced against equipment cost and safety.

2. Moderate temperature: Low temperature favours the exothermic forward reaction but gives a slow rate. About 400–450°C is a practical compromise.

3. Iron catalyst: It increases both rates and shortens the time to equilibrium without changing equilibrium yield.

4. Remove ammonia: Cool the product stream to condense NH_3, shifting further reaction towards products.

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5. Recycle reactants: Return unreacted N_2 and H_2 to the converter and maintain a suitable N_2:H_2 feed ratio.

These measures combine high equilibrium yield, practical reaction rate and efficient use of raw materials.

Points to Remember

Dynamic equilibrium: forward rate = reverse rate ≠ 0; macroscopic concentrations remain constant.

Equilibrium does not mean equal concentrations, complete reaction or stopped molecular activity.

Adding reactant or removing product shifts equilibrium forward; adding product or removing reactant shifts it backward.

For gases, higher pressure favours the side with fewer gas moles; if $\Delta n_{gas} = 0$, pressure does not change position.

Increasing temperature favours the endothermic direction and changes K; decreasing temperature favours the exothermic direction.

A catalyst changes the time to reach equilibrium, not K, ΔH or equilibrium composition.

Continuous product removal can drive a reversible reaction nearly to completion.

Haber yield: high pressure, moderate temperature, iron catalyst, NH_3 removal and reactant recycling.

MCQ key at a glance

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	a	2	d	3	a	4	b	5	c
6	c	7	b	8	a	9	b	10	a

How to attempt this chapter in the paper

• Write the balanced equation and physical states before predicting pressure effects; count gas moles only.
• State both the disturbance and the resulting direction; never say “equilibrium moves right” without identifying products.
• Do not confuse equilibrium position with rate: catalysts change rate, temperature changes both rate and K.
• For maximum industrial yield, explain the compromise between thermodynamics, rate, safety and cost.

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